Online Appendix: "The price of geoeconomic dependence: Sanction intensity and corporate credit spreads"

Benedikt Kagerer[1]

[1]Faculty of Economics, University of Cambridge. E-mail address: bmck3@cam.ac.uk.

Appendix Appendix A: Sample details

Appendix A.1 Bond sample

Table Appendix A.1: Summary statistics - bond sample

	P25	Median	P75	Mean	SD	N	Firms	Bonds
Issuance amount	400 000 000	500 000 000	750 000 000	578 160 783	333 568 002	1562575	366	1364
Duration	4.04	6.07	8.52	6.70	3.78	1562575	366	1364
Excess Bond Premium	-0.22	-0.04	0.17	0.01	1.41	1562575	366	1364
Default risk component	0.99	1.30	1.94	2.13	4.13	1562575	366	1364
5y PD	0.01	0.02	0.03	0.03	0.04	1562575	366	1364
Spread	0.93	1.30	1.98	2.13	4.40	1562575	366	1364
Yield	0.92	2.35	3.84	2.94	4.60	1562575	366	1364

Notes: Summary statistics for the bond characteristics of the bonds included in the complete sample from January 2014 to April 2024.

Table Appendix A.2: Bond market coverage

Year	Firms	Bonds	Average daily volume
2014	71	114	45 338 343 096
2015	97	164	72 787 643 409
2016	125	245	104 402 202 947
2017	164	340	158 889 104 123
2018	193	439	219 029 383 518
2019	245	615	295 310 037 217
2020	287	840	429 001 956 251
2021	330	1025	550 384 095 336
2022	343	1183	644 456 186 743
2023	358	1339	746 166 580 135
2024	353	1333	781 550 291 331

Notes: Coverage of firms and bonds over the complete sample period. Due to the limited data availability in the early years, all relevant bond market results are confirmed for the sub-period starting in 2022.

Further details on the bond selection and aggregation

- restrict the sample to issuance volumes above 1 million euros
- restrict the sample to senior-unsecured bonds
- remove bonds with less than 100 days of data
- remove bonds with yields above 100 percent
- keep bonds for which PD measures are available
- remove bonds with a negative spread
- remove stale priced bonds for which price remains unchanged over 20 days
- compute spreads by matching corporate bond yields with interest rates on German federal securities estimated for maturities of 6 months, 1 year, 2 years, up to 30 years by the Bundesbank
- industry and country aggregation: Compute aggregates if at least three different firms have bonds outstanding on each day

Appendix A.2 Structural VAR and firm data

Table Appendix A.3: Summary statistics - SVAR

	Mean	SD	P25	Median	P75
Spread	2.0798	0.36031	1.8094	2.024	2.2632
Default risk component	2.0958	0.17391	1.9569	2.0651	2.2107
EBP	-0.016046	0.24696	-0.1925	-0.051314	0.098335
Stoxx Europe 600	393.4004	46.8646	357.445	385.93	432.24
GPRD	116.435	53.2421	81.4733	106.857	140.8747
Gas TTF	33.7672	39.4071	15.38	19.99	28.4
VIXIE	41.3811	19.8446	29.6788	36.3907	46.9057

Notes: Summary statistics of the financial and news variables included in the SVAR estimation before the logarithmic transformation is applied to the Stoxx Europe 600, the GPRD, the Gas TTF, and the VIXIE series. Complete sample from January 2014 to April 2024.

Table Appendix A.4: Summary statistics - Firm financials

	Mean	SD	P25	Median	P75	N
Investment	0.015566	0.27926	-0.01375	0.007088	0.035536	2911
PPE (net, bn)	14.28687	35.0587	2.151	5.628	13.78	2911
Total Assets (bn)	48.47657	91.5569	10.3922	22.13814	50.0625	2911
Leverage	0.97544	23.68403	0.579405	0.995434	1.467415	2911
EBIT (mn)	702.034	1764.653	110.859	297.386	797.500	2880
Interest expense (mn)	-142.527	380.507	-114.371	-48.728	-23.180	2856
Spread	2.108984	4.386772	1.033244	1.378178	2.051838	2911

Notes: Summary statistics of the financial variables of bond issuing firms. Leverage is defined as total debt over total equity, investment as the log-difference in net property plant and equipment (PPE), and the spread is calculated as the unweighted average spread over all outstanding bonds per firm and quarter. Quarterly data between 2014 Q1 and 2024 Q1.

Table Appendix A.5: Firms with high Russia exposure

	Firms
High RU risk:	ASML HOLDING NV, BAYWA AG, CARLSBERG BREWERIES A/S, EIFFAGE SA, ELECTROLUX AB, ENEL SPA, ERAMET SA, FERROVIAL SE, FRAPORT FRANKFURT AIRPORT SERVICES, FRESENIUS SE & CO KGAA, HELLA GMBH & CO KGAA, HOCHTIEF AG, IMERYS SA, KEMIRA OYJ, METRO AG, NOKIA CORP, NOKIAN TYRES PLC, NV BEKAERT SA, PROSUS NV, QUADIENT SA, RENAULT SA, RWE AG, SACYR SA, SNAM SPA, SPIE SA, STORA ENSO OYJ, TERNA RETE ELETTRICA NAZIONALE SPA, UBISOFT ENTERTAINMENT SA, VEOLIA ENVIRONNEMENT SA, VINCI SA, VIVENDI SE, VOESTALPINE AG, WEBUILD SPA, YIT OYJ

Notes: This table lists firms classified as having a high Russia dependence, measured by the earnings call index of Hassan et al. (2024). Firms are classified as having a high Russia exposure if their Russia-specific risk index is above the 80th percentile on average in the years 2019 and 2020 (the last two available years before the full-scale invasion of Ukraine).

Appendix A.3 Short-term business statistics

Table Appendix A.6: Overview variables - Business statistics and country controls

Variable	Definition	Period	Source	Level
Bankruptcy	Total number of legal units filed for bankruptcy - Index	2015M1 - 2023M12	Eurostat - STBS	country-sector
Bankruptcy (quarterly)	Total number of legal units filed for bankruptcy - Index	2015M1 - 2023M12	Eurostat - STBS	country-sector
CPI	Consumer Price Index - Index	2014M1 - 2023M12	IMF - IFS	country
FX	Nominal effective exchange rate - Index	2014M1 - 2023M12	IMF - IFS	country
IP	Industrial Production - Index	2014M1 - 2023M12	IMF - IFS	country
10-y yield	Yield on 10-year government bonds	2014M1 - 2023M12	Datastream	country

Notes: Overview of the definition, coverage, and source of the variables included in the local projection estimations.

Table Appendix A.7: Summary statistics - Business statistics and country controls

variable	Mean	SD	P25	Median	P75	N	Countries
Bankruptcy	150.8543	79.99746	97.5	136.2	187.5	5521	9
Bankruptcy (qtr)	128.6294	69.15154	81.8	116.7	160.6	3249	18
Δ IP	0.087904	10.66736	-6.38783	0.103093	6.890665	5521	9
Δ 10-y yield	0.000697	0.154251	-0.1	-0.02	0.08	5521	9
Δ CPI	0.169855	0.524796	-0.10918	0.116138	0.42886	5521	9
Δ FX	0.103612	0.691957	-0.29153	0.016348	0.482246	5521	9

Notes: Summary statistics of the business statistics and country controls included in the local projection estimations.

Appendix Appendix B: Additional information - sanction shock identification

Appendix B.1 Further information on announcements and timeline of sanction measures

The EU imposed its first sanctions against Russia in response to the Russian annexation of Crimea in early 2014. These sanctions included travel bans and asset freezes of individuals involved in the Russian military operation. As Russia further escalated the situation through its military presence in the Ukrainian Eastern provinces of Donetsk and Luhansk, the US imposed sectoral sanctions, which according to the White House were "by far the most comprehensive sanctions applied to Russia since the end of the Cold War" *. Initially, the EU refrained from comparable measures. Only after the shooting down of flight MH17 by pro-separatist forces supported by Russia in July 2014, EU leaders decided to impose broader sectoral sanctions. The EU followed the US in implementing financial sanctions, imposed an arms embargo, and banned the export of dual-use products. No sanctions were imposed on the energy sector at the time by the EU. Russia responded to western sanctions with an import stop for agricultural products from countries imposing sanctions on Russia in an effort to limit its western dependence. Contrary to what would later unfold during the full-scale invasion of Ukraine in 2022, Russia was not able to profit from rising energy prices after the annexation of Crimea in 2014. Falling oil prices, as the OPEC attempted to outprice US shale production, further weighed on Russian state finances. The US sanctions on the energy sector made a bailout of Russia's largest oil company Rosneft through the Russian central bank necessary. This bailout led to a 30% collapse of the ruble in a single day with major repercussions for the Russian economy. The economic impact for the EU of sanctions imposed against Russia in 2014 and the beginning of 2015 was mild, given that crucial links, in particular in the energy sector, were not touched and Russian countersanctions were only weakly enforced. While sanctions were extended and tightened over the years 2015 to 2018, significant new measures were only taken after the attempted assassination of former agent Sergei Skripal and his daughter on British soil followed by an expulsion of diplomats on both sides.

On February 23, 2022, the day before the Russian full-scale invasion of Ukraine, the EU imposed the first sanction package as a response to the Russian decision "to proceed with the recognition of the non-government controlled areas of Donetsk and Luhansk oblasts in Ukraine as independent entities, and the ensuing decision to

*The White House Press Office

send Russian troops into these areas" [†]. Over the following two years, another 12 sanction packages would be announced by the EU. This time, far-reaching financial and economic measures were taken, including a ban of 10 Russian banks from the international payment system SWIFT as well as an import ban on the energy sector. The imposed sanctions include six main areas: Energy, Finance, Transport, Trade, Services, and Defense and Technology. Import restrictions among others include oil and coal, materials such as iron and steel, wood, paper and aluminium, as well as gold, diamonds, spirits, and seafood. Export restrictions were imposed on electronic components, chemicals, luxury goods including vehicles, and machinery components. In total, 54% of pre-war exports from the EU and 58% of pre-war imports in the EU from Russia fall under these measures [‡]. While no EU sanctions were imposed on gas imports, the stop of pipeline gas imports through Northstream 1 and 2 and the Yamal pipeline meant a substantial drop in total gas imports from Russia. Germany in particular, having been the main importer of Russian gas before the war, stopped importing gas directly from Russia. Russia responded to Western sanctions with FX restrictions for foreign banks, restrictions on dividend payments to foreigners, and export restrictions of selected goods among others. A full timeline of these and other measures taken by both parties is given in Online Appendix B, Table Appendix B.1.

[†] European Council
[‡] European Commission

Table Appendix B.1: List of announcement days of restrictive measures by the EU and Russia

Date	Event	Announcement by
06.03.2014	Extraordinary meeting EU heads of state; Start preparation individual sanctions	EU
17.03.2014	Introduction of a first set of restrictive measures against 21 Russian and Ukrainian officials	EU
20.03.2014	EU leader request preparation of broader economic sanctions; futher names added to individual sanctions	EU
12.05.2014	Agreement on new set of sanctions	EU
23.06.2014	Import ban on goods from Crimea	EU
16.07.2014	Meeting European Council: 6 additional restrictive measures against Russia	EU
18.07.2014	Ukraine crisis: EU broadens remit of sanctions	EU
22.07.2014	Council demands sanctions in four sectors after downing of MH17	EU
25.07.2014	Further EU sanctions over situation in Eastern Ukraine	EU
29.07.2014	Adoption of economic sanctions against Russia	EU
06.08.2014	Russia bans the import of foodstuffs from sanctioning countries	RU
07.08.2014	Russia defines list of sanctioned products and countries	RU
30.08.2014	European Council asks for preparation of new economic sanctions	EU
12.09.2014	Further economic sanctions on Russia	EU
17.11.2014	EU ministers aks for proposal on further economic sanctions	EU
28.11.2014	EU strengthens sanctions against separatists in Eastern Ukraine	EU
12.02.2015	Minsk II	EU/RU
16.02.2015	EU strengthens sanctions against separatists in Eastern Ukraine	EU
30.03.2018	Russia orders diplomats out (Skripal)	RU
23.02.2022	First package of sanctions against Russia	EU
24.02.2022	Russian full-scale invasion of Ukraine starts; EU leaders agree on further sanctions	EU/RU
25.02.2022	Second package of sanctions in response to Russia's invasion of Ukraine	EU
28.02.2022	Third package of sanctions in response to Russia's invasion of Ukraine / FX restrictions to balance ruble (Decree 79) / Ban EU air carriers	EU/RU
01.03.2022	Restrictions cash withdrawal (Decree 81)	RU
02.03.2022	SWIFT ban Russian banks	EU
04.03.2022	Profit payments to foreigners (Decree 254), Sanctions on individuals (Decree 30-FZ)	RU
06.03.2022	Procedure permits for transactions (Decree 295)	RU
08.03.2022	Export ban (Decree 100, Government order 311)	RU
14.03.2022	Ban on transactions of RU insurance corporations with foreign (Law 55-FZ)	RU
15.03.2022	Fourth package of sanctions in response to Russia's invasion of Ukraine	EU
18.03.2022	Additional restrictions of FX and transactions (Decree 126)	RU
31.03.2022	Payment gas in rubles (Decree 172)	RU
01.04.2022	Foreign banks not allowed to buy other currencies in Russia (Decision "On Establishment of the Amount of Certain Transactions of Residents and Non-residents")	RU
08.04.2022	Fifth package of sanctions in response to Russia's invasion of Ukraine	EU
16.04.2022	Ban on issuance and trade of depositary receipts outside Russia (114-FZ)	RU
03.05.2022	Trade with sanctioned individuals (Decree 252)	RU
27.05.2022	Payments for IP via special accounts (Decree 322)	RU
31.05.2022	European Council agrees on sixth sanctions package	EU
06.10.2022	Eighth package of sanctions in response to Russia's invasion of Ukraine	EU
16.12.2022	Ninth package of sanctions in response to Russia's invasion of Ukraine	EU
25.02.2023	Tenth package of sanctions in response to Russia's invasion of Ukraine	EU
23.06.2023	Eleventh package of sanctions in response to Russia's invasion of Ukraine	EU
18.12.2023	Twelfth package of sanctions in response to Russia's invasion of Ukraine	EU
23.02.2024	EU adopts 13th package of sanctions against Russia	EU

Table Appendix B.2: Official sanction announcements and sanction intensity in 2014 and 2022

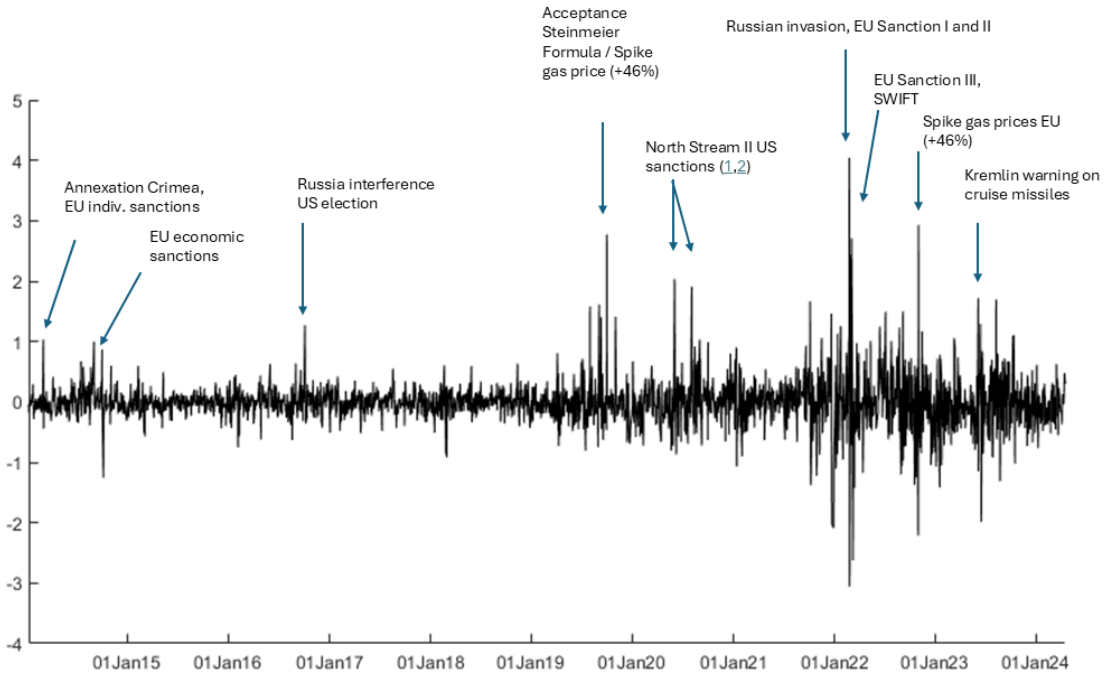
	(1)	(2)	(3)	(4)	(5)	(6)
	2014			2022		
	Sanction news	Δ Sanction news	σ^2_{TTF}	Sanction news	Δ Sanction news	σ^2_{TTF}
announcement	1.309***	0.293***	0.001	1.427***	0.010	0.028**
	(0.182)	(0.095)	(0.001)	(0.186)	(0.063)	(0.013)
Observations	244	243	243	242	241	242
R^2	0.074	0.021	0.030	0.238	0.000	0.106

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: A logarithmic transformation is applied to the news indices and gas prices.

Figure Appendix B.1: Shock series and major conflict events



Notes: Daily sanction shock series extracted from the structural VAR including credit spreads, and the logarithmic transformation of the Stoxx Europe, the GPRD, the Gas TTF, and the VIXIE. Individual spikes are connected to conflict events.

Appendix B.2 The pivotal role of gas prices for sanction shock identification

The previous analysis highlighted the importance of gas prices as a highly responsive series to EU-Russia conflict for the identification of the sanction shock. The pivotal role of gas prices in the identification approach is also reflected in the estimates of the variance-covariance matrices of the VAR reduced form errors under the two regimes. Both the variance and the covariances between gas prices and the other financial and news variables in the model change strongly between announcement and non-announcement days, with substantially stronger co-movement on announcement days (Tables Appendix B.3 and Appendix B.4).

Table Appendix B.3: Estimate variance-covariance matrix of reduced form errors - non-announcement days

$\hat{\Sigma}_0$	Spread	Estoxx	GPRD	VIXIE	TTF Gas
Spread	0.028	-0.002	-0.001	0.004	0.000
Estoxx		0.008	0.003	-0.010	0.000
GPRD			0.651	-0.005	0.001
VIXIE				0.026	0.000
TTF Gas					0.003

Table Appendix B.4: Estimate variance-covariance matrix of reduced form errors - announcement days

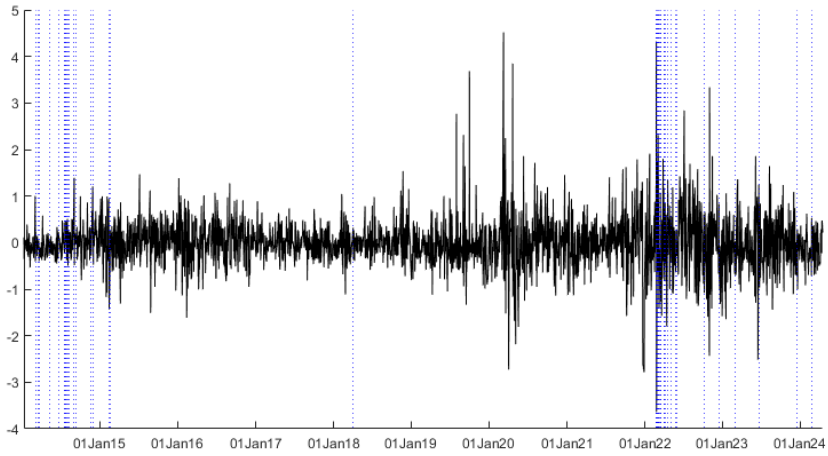
$\hat{\Sigma}_1$	Spread	Estoxx	GPRD	VIXIE	TTF Gas
Spread	0.029	-0.003	-0.004	0.005	0.011
Estoxx		0.010	0.004	-0.015	-0.011
GPRD			0.655	0.002	0.035
VIXIE				0.042	0.020
TTF Gas					0.030

Notes: The variance-covariance matrices are calculated as $\hat{\Sigma}_i = \frac{1}{\#T_i} \sum_{t \in T_i} \hat{u}_i \hat{u}_i'$ for sub-samples $i = 0, 1$. The variance-covariances are estimated over the complete sample of daily data between January 2014 and April 2024.

One potential concern regarding the identification of the sanction shock is that factors, unrelated to the EU-Russia conflict, could impact the EU commodity prices on the selected event-days. To rule out this possibility, I linearly regress the TTF front-month gas prices on the Brent oil price (and a constant) to separate comovement in EU energy prices from the orthogonal gas price-specific component. Before the full-scale invasion, over 40% of the EU's gas imports and 25% of its oil imports came from Russia. Moreover, the EU relied heavily on its pipeline infrastructure for cheap access to natural gas from Russia. The orthogonal component of gas prices is, thus, expected to capture the conflict effects, beyond what is already reflected in oil prices.

I replace the gas price series in the structural VAR estimation by the gas price component orthogonal to oil prices and re-estimate the model under precisely the same assumptions as in the baseline specification. Again, we reject the assumption of equal variance-covariance matrices on announcement days and non-announcement days over the full sample period (p-value: 0.01). We again do not reject the null hypothesis of a single shock only on announcement days (p-value: 0.83). Also, the shock series resembles the series obtained in the baseline model specification.

Figure Appendix B.2: Sanction shock series - Gas price component orthogonal to oil prices

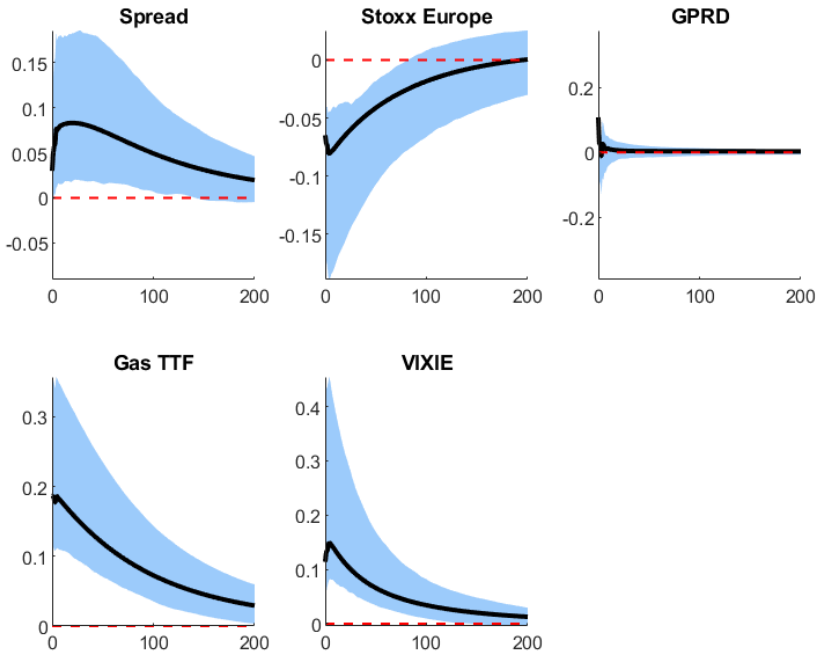


Notes: Daily sanction shock series extracted from the structural VAR including credit spreads, and the logarithmic transformation of the Stoxx Europe, the GPRD, the component of Gas TTF orthogonal to the Brent oil price, and the VIXIE. Dotted lines correspond to the 44 (counter)sanction announcement days selected to identify the shock.

Impulse response functions to a unitary sanction shock, as estimated in the model including the gas price component unrelated to oil price movements, confirm the findings of the baseline model specification. Even though the effect of the sanction shock in this setting is somewhat dampened, significant effects on the credit spreads are again found especially for the sample of intensified conflict starting in 2022 (Figures Appendix B.3, Appendix B.4).

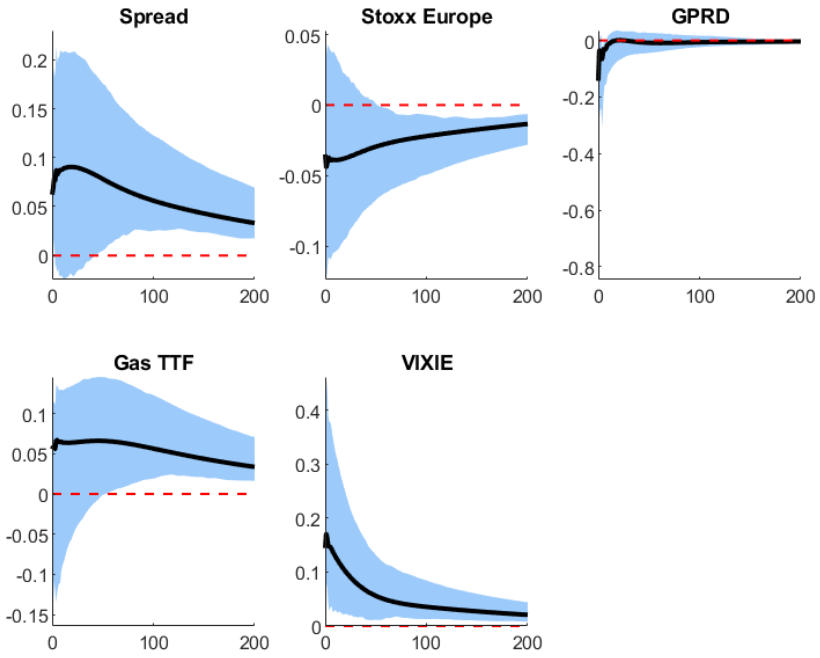
If we replace the EU gas price series by the CBOE Crude Oil Volatility Index, reflecting uncertainty in global oil prices, and re-estimate the structural VAR, the change in the variance-covariance matrix between announcement and non-announcement days is not sufficient to identify the shock (p-value: 0.68). This confirms that general energy uncertainty cannot explain the observed results.

Figure Appendix B.3: Effect of a sanction shock on credit spreads 2014 to 2024 - Gas price component orthogonal to oil prices



Notes: Impulse response function to a unitary adverse sanction shock together with the 90% bootstrapped standard errors. A logarithmic transformation is applied to the Stoxx Europe, GPRD, Gas TTF, and VIXIE. Only the component of the Gas TTF orthogonal to Brent oil prices is included. All variables are standardised to have a zero mean and a unit variance. Daily data between January 2, 2014 and April 15, 2024.

Figure Appendix B.4: Effect of a sanction shock on credit spreads 2022 to 2024 - Gas price component orthogonal to oil prices



Notes: Impulse response function to a unitary adverse sanction shock together with the 90% bootstrapped standard errors. A logarithmic transformation is applied to the Stoxx Europe, GPRD, Gas TTF, and VIXIE. Only the component of the Gas TTF orthogonal to Brent oil prices is included. All variables are standardised to have a zero mean and a unit variance. Daily data between January 4, 2022 and April 15, 2024.

Appendix B.3 Robustness: Excluding days with economic events and data releases

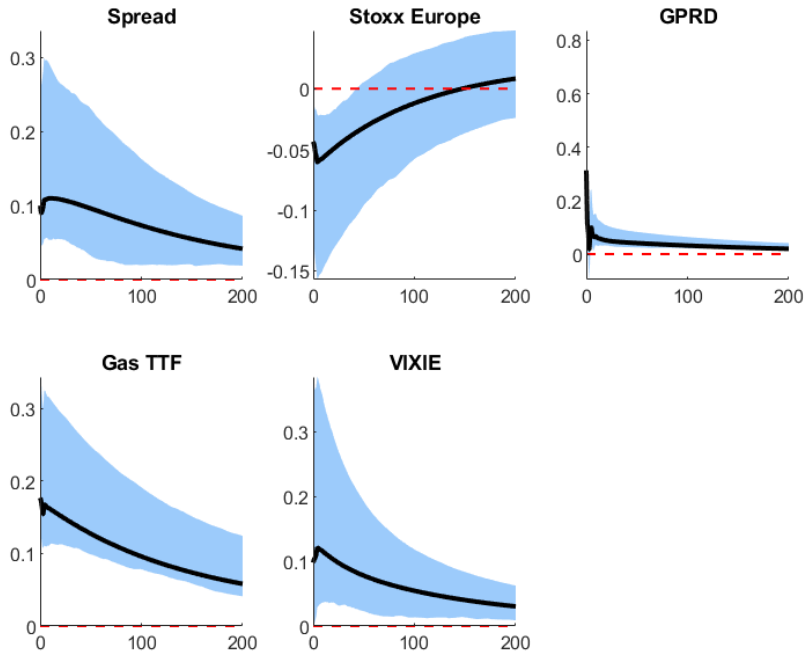
To exclude the possibility of other events contaminating the selected event days, I proceed as follows: In a first step, I collect economic events and data releases coinciding with sanction announcement days from the Bloomberg Eurozone calendar. Economic events coincide with sanction announcements on eight days, which I exclude in a robustness check. Results remain extremely robust (Figure Appendix B.5). Data releases are substantially more frequent (releases happen on 32 of the 44 announcement days) such that overlapping days cannot simply be excluded. Therefore, for each announcement day with economic releases, I assess the relevance of the release using two different approaches. First, I use the Bloomberg relevance score and exclude event days for which a release with a relevance above 80 was made. Using this logic 12 announcement days are removed. Secondly, I construct a metric for the importance of the economic data release relative to the sanction announcement. I compare the number of articles on an overlapping day that contain the keywords of the corresponding release (e.g. "CPI", "PMI", "unemployment rate", etc. and one of "Eurozone", "euro area", "Europe", "EU") with the number of articles mentioning "sanctions", "Russia", and "Ukraine". This gives an indication of the relative signal strength the VAR picks up from the sanction intensity shock and from a possible other economic shock on such an announcement day. I exclude articles for which the share of release articles relative to sanction articles exceeds 0.25. This strategy removes 14 announcement days. Results are extremely robust to both approaches (Figures Appendix B.6 and Appendix B.7). Excluding days with a high relative importance of economic data releases even strengthens the effects somewhat. This confirms that the results are not driven by alternative macro events and indeed capture effects of a sanction intensity shock.

Table Appendix B.5: Test identification assumptions (p-values)

Sample	H0: $\Sigma_1 = \Sigma_0$	H0: $\Sigma_1 - \Sigma_0 = b_1 b_1'$
Excluding events	0.000	0.846
Excluding releases relative	0.000	0.854
Excluding releases absolute	0.000	0.880

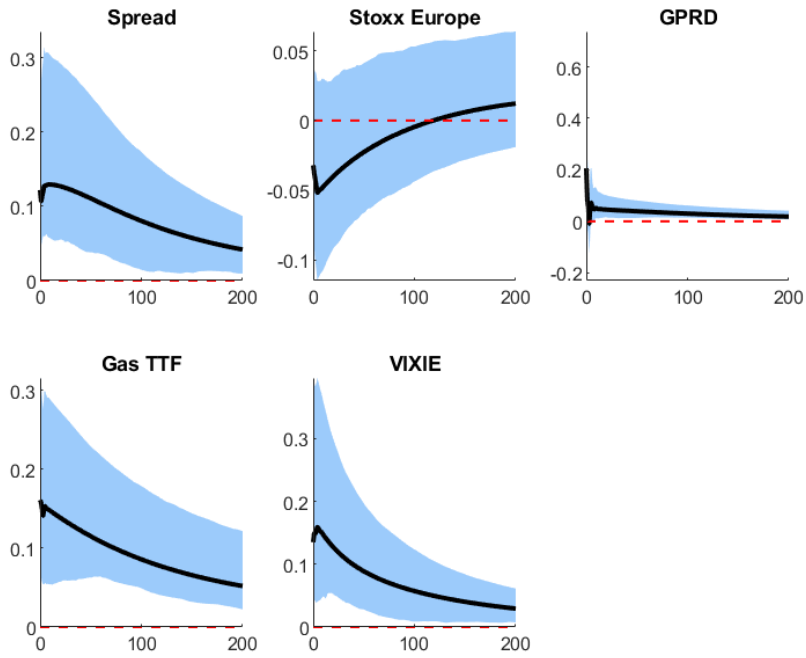
Notes: P-values derived from the bootstrapped distribution of the Wald test statistics given in Equation 12 (left column) and Equation 10 (right column). All observations are used in all samples to compute the variance-covariance matrices. A moving-block bootstrap of Jentsch and Lunsford (2019) is performed with 1000 repetitions. Following the authors, the block length is set at $l = 5 * T^{\frac{1}{4}}$ for each sample. Additional information on the implementation of the bootstrap is given by Boer et al. (2023).

Figure Appendix B.5: Effect of a sanction shock on credit spreads, 2014 to 2024, excluding economic events



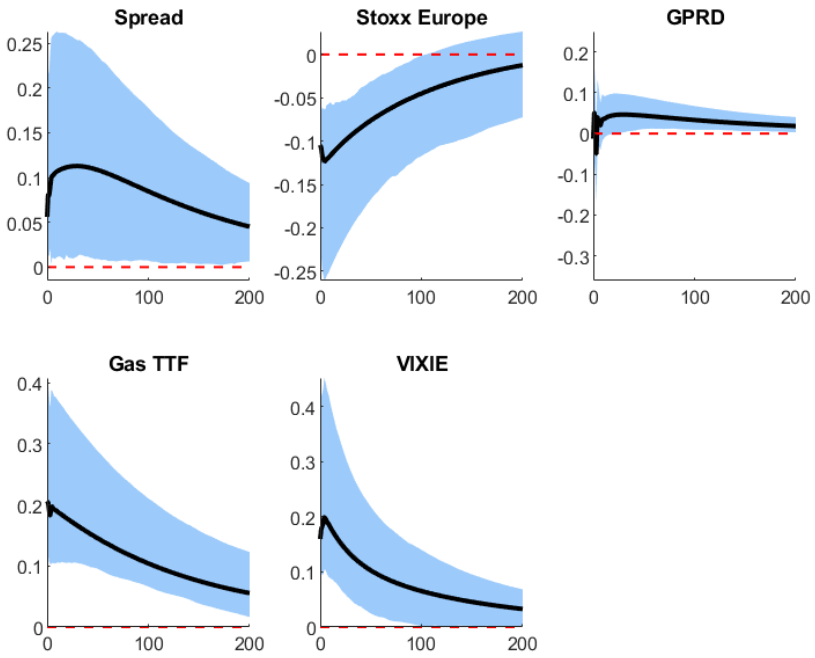
Notes: Impulse response function to a unitary adverse sanction shock together with the 90% bootstrapped standard errors over a 200-day horizon. Sanction announcement days coinciding with economic events are excluded. A logarithmic transformation is applied to the Stoxx Europe, GPRD, Gas TTF, and VIXIE. All variables are standardised to have a zero mean and a unit variance. Daily data between January 2, 2014 and April 15, 2024.

Figure Appendix B.6: Effect of a sanction shock on credit spreads, 2014 to 2024, excluding economic release days (relative)



Notes: Impulse response function to a unitary adverse sanction shock together with the 90% bootstrapped standard errors over a 200-day horizon. Sanction announcement days on which newspaper coverage of economic data releases exceed 0.25 the coverage of sanctions are excluded. A logarithmic transformation is applied to the Stoxx Europe, GPRD, Gas TTF, and VIXIE. All variables are standardised to have a zero mean and a unit variance. Daily data between January 2, 2014 and April 15, 2024.

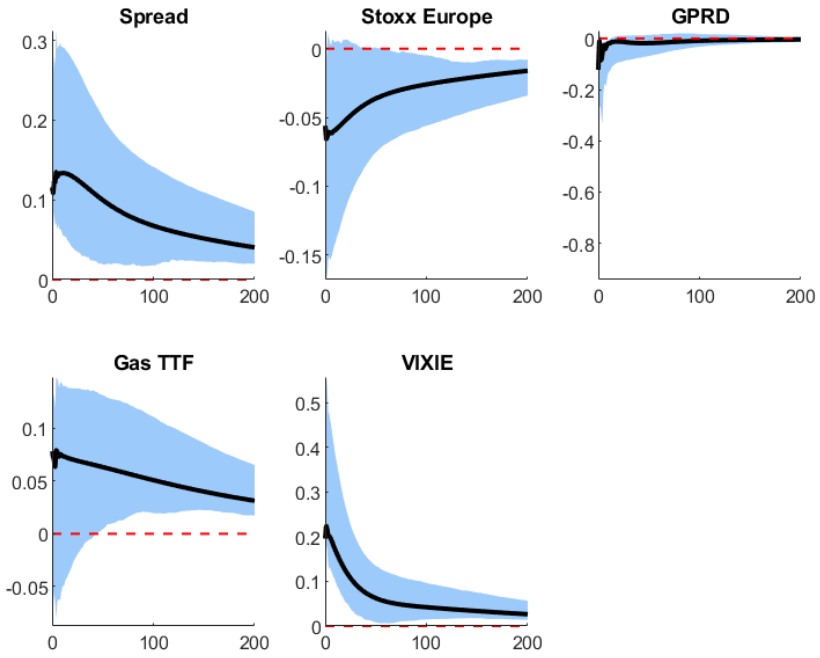
Figure Appendix B.7: Effect of a sanction shock on credit spreads, 2014 to 2024, excluding economic release days (absolute)



Notes: Impulse response function to a unitary adverse sanction shock together with the 90% bootstrapped standard errors over a 200-day horizon. Sanction announcement days on which economic data releases with a Bloomberg relevance score above 80 are excluded. A logarithmic transformation is applied to the Stoxx Europe, GPRD, Gas TTF, and VIXIE. All variables are standardised to have a zero mean and a unit variance. Daily data between January 2, 2014 and April 15, 2024.

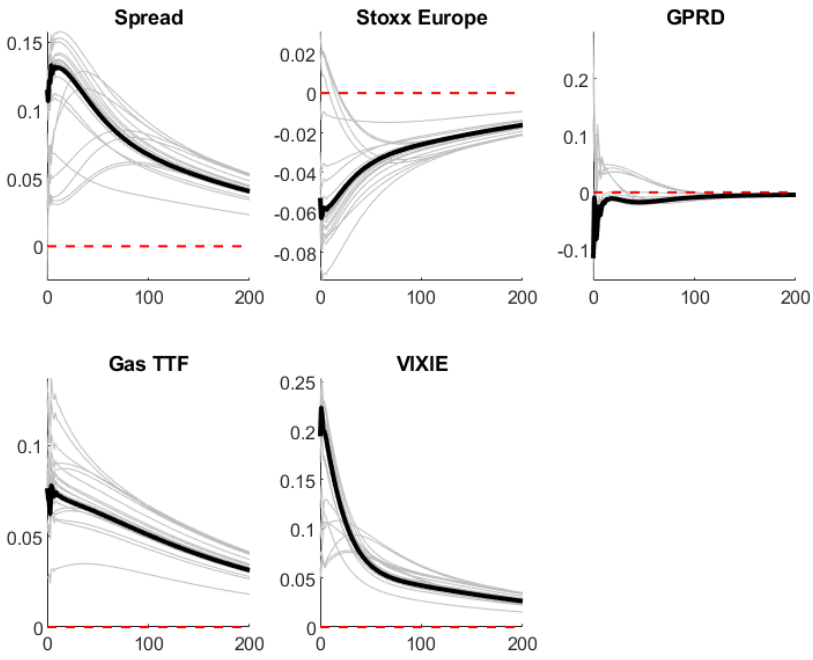
However, the result, might still be contaminated by unscheduled economic news that overlap with sanction announcements. To rule out this option, I collect the top Bloomberg Global Macro news headlines marked with high importance on sanction announcement days. Given that the results are driven by the post-2022 period, I limit this exercise to the shorter post-2022 sample. Table Appendix B.6 gives the collection of the top three headlines on each announcement day that are not directly discussing the EU-Russia conflict. In a second step, I apply judgement on whether these news are i) sufficiently material and ii) whether they apply to EU firms. I exclude 3 dates from the announcement sample that are both judged sufficiently material and that apply directly to the EU. Results survive the exclusion of these days virtually unchanged (Figure Appendix B.8). To ensure that no single other overlapping event explains the results, I further iteratively remove all individual announcement days from the sample. Reassuringly, all specifications show a strong tightening of corporate financial conditions in response to the sanction intensity shock (Figure Appendix B.9).

Figure Appendix B.8: Effect of a sanction shock on credit spreads, 2022 to 2024 excluding overlapping news



Notes: Impulse response function to a unitary adverse sanction shock together with the 90% bootstrapped standard errors over a 200-day horizon. Announcement days exclude March 8, 2022, April 8, 2022, and 27 February, 2023, due to overlapping news. A logarithmic transformation is applied to the Stoxx Europe, GPRD, Gas TTF, and VIXIE. All variables are standardised to have a zero mean and a unit variance. Daily data between January 4, 2022 and April 15, 2024.

Figure Appendix B.9: Effect of a sanction shock on credit spreads, 2022 to 2024, iteratively removing each of the announcement days



Notes: Impulse response function to a unitary adverse sanction shock calculated by iteratively removing individual announcement days. A logarithmic transformation is applied to the Stoxx Europe, GPRD, Gas TTF, and VIXIE. All variables are standardised to have a zero mean and a unit variance. Daily data between January 4, 2022 and April 15, 2024.

Table Appendix B.6: News and relevant events

Date	News 1	News 2	News 3	Rel.	Europe
23.02.2022	Fed's Daly Backs March Move Absent Big 'Negative Surprises'	New Zealand Raises Rates a Third Time, Signals Higher Peak (1)	Soybeans Soar to 9-Year High With South America Supply in Doubt		
24.02.2022	U.S. New-Home Sales Decline for First Time in Three Months	Morgan Stanley Discloses U.S. Probe Into Block-Trading Business	*U.S. WEEKLY JOBLESS CLAIMS AT 232,000 LAST WEEK; EST. 235,000		
25.02.2022	CDC Loosens Guidelines for Mask-Wearing as Omicron Wave Recedes	Hedge Funds Shorting Stocks Face More Disclosure Rules From SEC	*U.S. JAN. PERSONAL SPENDING RISES 2.1% M/M; EST. 1.6%		
28.02.2022	Treasuries Extend Gains, Two-Year Yield Falls 10 Bps to 1.47%	*TOYOTA TO SUSPEND JAPAN PLANTS ON SUPPLIER CYBER ATTACK: NIKKEI	Fed's Bostic Still Favors 25 Basis-Point Move at March Meeting		
01.03.2022	Powell Seeks to Reassure Lawmakers Fed Will Curb Hot Inflation	Traders Abandon Bets on a Half-Point Fed Rate Hike in March	Manufacturing Growth in U.S. Firms on Stronger Orders, Output		
02.03.2022	*POWELL: IF INFLATION STAYS HOT, WE COULD HIKE 50 BP AT A MTG	Bullard Calls for 'Rapid Withdrawal' of Fed Policy Accommodation	Biden Says His 'Top Priority' Is Getting Prices Under Control	x	
04.03.2022	*U.S. FEB. UNEMPLOYMENT RATE FALLS TO 3.8% VS 4.0%	*TREASURY FIVE-YEAR YIELD DROPS 10BPS TO 1.63% ON HAVEN DEMAND	*EURO EXTENDS DECLINE BELOW $1.10 FOR FIRST TIME SINCE MAY 2020		x

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Table Appendix B.6 – continued from previous page

Date	News 1	News 2	News 3	Rel.	Europe
06.03.2022	*EURO DROPS BELOW PARITY WITH SWISS FRANC, FIRST TIME SINCE 2015	U.K. House Prices Surge at Strongest Annual Pace Since 2007	Bond Market's U.S. Inflation Expectation Hits Highest Since 2005		x
08.03.2022	Nickel Resumes Advance After Manic Monday That Smashed Records	China Asks Banks to Report Derivatives Exposure as Markets Reel	**EU to Consider Massive Joint Bond Sales to Fund Energy, Defense**	x	x
14.03.2022	Foxconn Halts IPhone Shenzhen Site Due to Covid Lockdown	Fed Traders Move to Fully Price In Seven Standard Hikes for 2022	China Locks Down Tech Hub of Shenzhen as Covid Outbreak Spreads		
15.03.2022	Wild Swings Rule China Stocks as Bonds, Yuan Also Slide	U.S. Producer Prices Rise 10%, Setting Stage for Fed Rate Hike	Saudi Weighs Using Yuan Instead of USD for China Oil Sales: DJ		
18.03.2022	China Weighs Tencent Payments Overhaul, New License Requirement	U.S. Feb. Existing-Home Sales Fell 7.2% to 6.02m, Below Est.	Putin Proposes Nabiullina for New Term as Central Bank Governor		
31.03.2022	U.S. Mortgage Rates Soar Again, Reaching Highest Since Late 2018	China Weighs Raising Billions to Rescue Troubled Financial Firms	U.S. Jobless Claims Rise More Than Forecast, But Level Still Low		
01.04.2022	China Weighs Giving U.S. Full Access to Audits of Most Firms	Fed's Evans Favors Quarter-Point Rate Hikes at Each 2022 Meeting	U.S. Hiring Stays Robust as Jobless Rate Falls, Wages Pick Up		
08.04.2022	Goldman's Hatzius Sees Chance Fed May Need to Hike Rates Past 4%	**ECB Is Crafting a Crisis Tool to Activate If Bond Yields Jump**	Goldman Traders Make $300 Million on Inflation in First Quarter	x	x

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Table Appendix B.6 – continued from previous page

Date	News 1	News 2	News 3	Rel.	Europe
16.04.2022	Mask Mandate for Planes, Trains Overturned by Federal Judge	U.S. Natural Gas Surges to 13-Year High on Global Supply Crunch	Fed's Bullard Says 75 Basis-Point Hike Could Be Option If Needed	x	
03.05.2022	U.S. March Factory Goods Orders Rise 2.2%; Est. Up 1.2%	*RBA RAISES CASH RATE TARGET 25 BASIS POINTS TO 0.35%	*FITCH CUTS CHINA 2022 GROWTH FORECAST AMID COVID-19 OUTBREAKS		
27.05.2022	China Opens Exchange Bond Markets to Overseas Investors: PBOC(1)	Firmer US Inflation-Adjusted Spending Underscores Solid Consumer	Beijing Adds More Cases as Anxiety Over China Economy Swells		
31.05.2022	Treasuries Sell Off Across the Curve as Cash Trading Resumes	Deutsche Bank's DWS Unit Raided Amid Allegations of Greenwashing	Yellen Says 'I Was Wrong' Last Year on the Path of US Inflation	x	
06.10.2022	US Inflation Unexpectedly Accelerates to 40-Year High of 8.6%	Covid Flares Anew in Shanghai, Sending Some Back to Lockdown (1)	Treasury Yields Hit Highest Since 2008 as Fed Path Repriced	x	
16.12.2022	China Hints at More Property Support as Economy Takes Priority	China-HK Border Will 'Fully Reopen' Next Month, SCMP Reports	Covid Unleashed in Beijing Shows Rest of China What Comes Next		
25.02.2023	ECB Must Make Sure Inflation Returns to 2%, Lagarde Tells HS	Australia Recession Risk Rises as RBA Seen Hiking More Than Fed	**ECB's Lagarde Sees 'Every Reason' for Half-Point March Hike: ET**	x	x
23.06.2023	Goldman Sachs Is Cutting About 125 of Its Managing Directors	Fed's Bostic Sees Inflation Heading Lower, Favors Holding Rates	*EURO-AREA JUNE MANUFAC-TURING PMI 43.6; FORECAST 44.8		x

Continued on next page

Table Appendix B.6 – continued from previous page

Date	News 1	News 2	News 3	Rel.	Europe
18.12.2023	Fed's Goolsbee Says Markets Front-Running FOMC on Rate Cuts	Fed Officials Add to Chorus Pushing Back Against Rate-Cut Bets	Macklem Sees Bank of Canada Cutting Rates Sometime in 2024		
23.02.2024	Goldman Pushes Back Bet on First Fed Interest-Rate Cut to June	Top Fed Officials Bolster Case for Patient Stance on Rate Cuts	Fed's Waller Urges Patience on Rate Cuts After Jump in Prices		

Appendix B.4 An alternative model with multiple shocks

The assumption of a single sanction intensity shock on announcement days can be relaxed to a setting in which all shocks in the model are allowed to change variance between announcement days and non-announcement days. In this setting, the restrictiveness of the identification assumptions can be evaluated. The covariance matrices on announcement days (Σ_1) and non-announcement days (Σ_0) can be decomposed into

$$\Sigma_0 = BB' \; and \; \Sigma_1 = B\Lambda B' \qquad \text{(Appendix B.1)}$$

where $Lambda = diag(\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5)$ contains the structural shock variances on announcement days and B the constant impact coefficients. As before, structural shock variances on non-announcement days are normalised to one. We can re-write $\Sigma_1\Sigma_0^{-1} = B\Lambda B' \, (BB')^{-1} = B\Lambda(B')^{-1}$, giving the eigendecomposition of the matrix $\Sigma_1\Sigma_0^{-1}$. Estimating this matrix, the coefficient vectors $\hat{b_i}$ and the variance shifters $\hat{\lambda_i}$ for $i = 1, ..., 5$ are obtained as the eigenvectors and eigenvalues, respectively (Lütkepohl et al. (2021), Boer et al. (2023)). The resulting variance shifts are reported in Table Appendix B.7 for the full sample period and in Table Appendix B.8 for the sub-period starting in 2022. These shifts let us investigate the two main assumptions in the partially identified model: 1) The presence of a shift in variance in the first shock and 2) no other shifts in the variances of the other shocks. The results provide clear evidence of a unique shock on sanction announcement days. The shock estimate for the first shock is with $\hat{\lambda_1} = 9.704$ for the complete sample period and with $\hat{\lambda_1} = 6.353$ for the post-2022 period, significantly larger than one on a 5% significance level. All other shocks in the model do not experience an increase in their variance on announcement days relative to non-announcement days at this significance level. For the post-2022 period, we see, while statistically insignificant, some increase in the variances of the second, third, and fourth shock. Figure Appendix B.10 shows the impulse response function of the dominating shock in this unrestricted model. We observe very similar dynamics as for the restricted model. This suggests that even if we don't rule out the possibility of other shocks changing on announcement days after 2022, the dominant shock behaves like the sanction intensity shock over the full sample period. These additional results confirm that the results in the paper are not driven by inappropriate assumptions and confirm the applicability of the restricted model used in the main part of the paper.

Table Appendix B.7: Relative variances on sanction announcement days

	$\hat{\lambda}_1$	$\hat{\lambda}_2$	$\hat{\lambda}_3$	$\hat{\lambda}_4$	$\hat{\lambda}_5$
2.5%	1.284	0.750	0.483	0.295	0.137
5%	1.456	0.820	0.544	0.336	0.173
16%	2.063	1.005	0.658	0.433	0.255
Estimate	9.704	1.367	1.064	0.915	0.481
84%	14.273	1.840	1.306	0.961	0.531
95%	17.658	2.247	1.491	1.110	0.610
97.5%	18.808	2.450	1.597	1.179	0.650

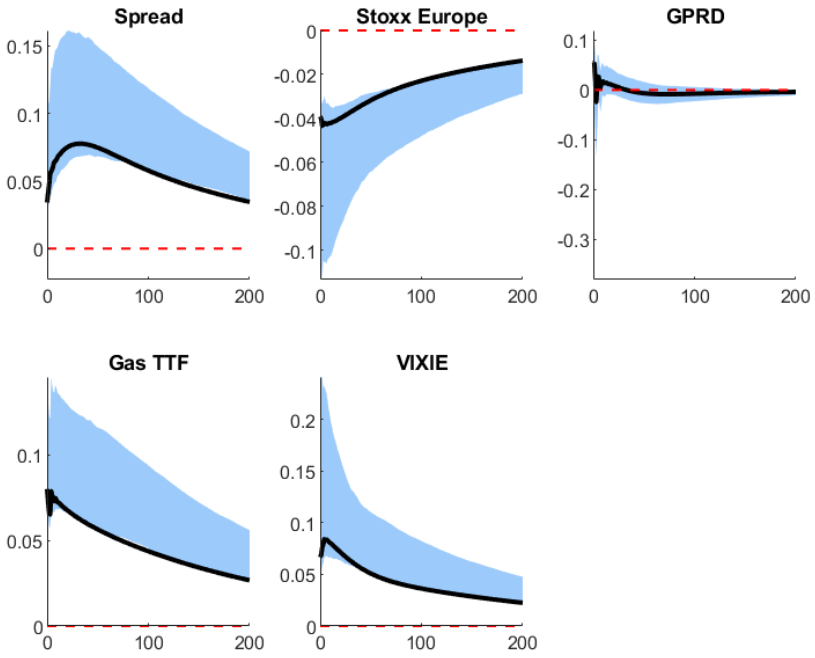
Notes: 68%, 90%, and 95% confidence intervals obtained from a moving-block bootstrap with 1000 repetitions. Full sample from 2014 to 2024.

Table Appendix B.8: Relative variances on sanction announcement days

	$\hat{\lambda}_1$	$\hat{\lambda}_2$	$\hat{\lambda}_3$	$\hat{\lambda}_4$	$\hat{\lambda}_5$
2.5%	1.531	0.867	0.400	0.154	0.021
5%	1.763	1.017	0.480	0.225	0.148
16%	2.541	1.399	0.743	0.381	0.148
Estimate	6.353	1.827	1.604	1.348	0.580
84%	8.347	2.492	1.847	1.287	0.590
95%	10.055	3.109	2.115	1.591	0.701
97.5%	11.458	3.486	2.253	1.708	0.754

Notes: 68%, 90%, and 95% confidence intervals obtained from a moving-block bootstrap with 1000 repetitions. Sub-sample from 2022 to 2024.

Figure Appendix B.10: Effect of a sanction shock on credit spreads 2022 to 2024 - Fully identified model



Notes: Impulse response function to a unitary adverse sanction shock together with the 90% bootstrapped standard errors. A logarithmic transformation is applied to the Stoxx Europe, GPRD, Gas TTF, and VIXIE. Only the component of the Gas TTF orthogonal to Brent oil prices is included. All variables are standardised to have a zero mean and a unit variance. Daily data between January 1, 2022 and April 15, 2024.

Appendix B.5 Alternative shock identification approach: Proxy-VAR

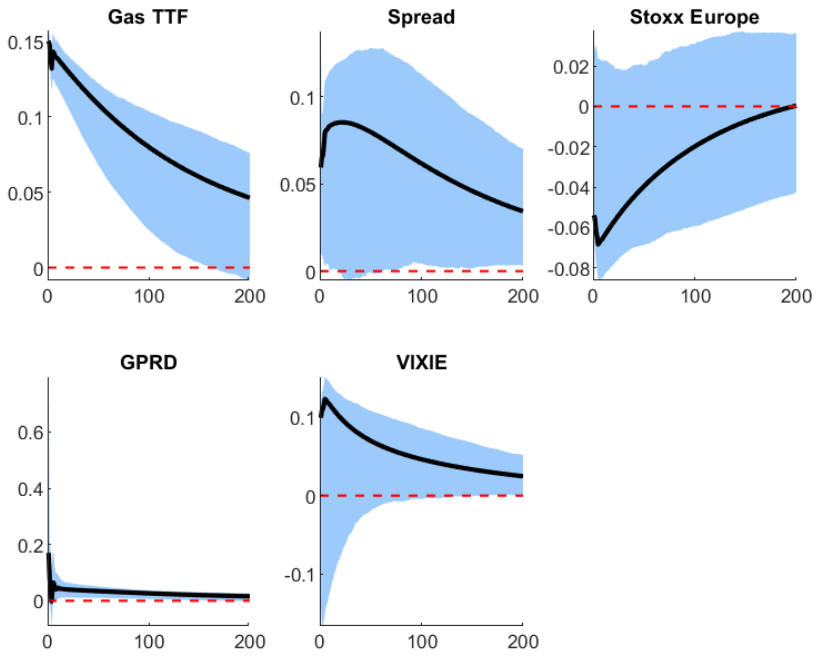
The shock identification presented in the main part of the paper follows a statistical approach exploiting variance changes to deduce shock intensity, with the caveat that assumptions on the structure of the variance changes are needed. An alternative approach, which also exploits the responsiveness of gas prices to sanction announcements, is a Proxy-VAR. This approach does not rely on assumptions regarding the structure of shock variances, instead it assumes exogeneity and relevance of the instrument:

$$E[z_t \epsilon_{1,t}] = \alpha \neq 0 \qquad \text{(Appendix B.2)}$$

$$E[z_t \epsilon_{2:n,t}] = 0 \qquad \text{(Appendix B.3)}$$

I follow the methodological approach of Stock and Watson (2012) and Mertens and Ravn (2013) and use an external proxy, in this case the sanction announcements, to identify sanction intensity in the VAR. Specifically, to increase instrument strength, I consider the change in gas prices on the announcement days. Assuming that gas price changes on announcement days capture sanction intensity and that no other shocks lead to price movements on such days, the resulting dynamics give the effect of a sanction intensity shock purely identified using gas prices on corporate credit spreads. I verify that Alessandri and Gazzani (2025) do not identify gas supply shocks that are not sanction-related on announcement days. When regressing the gas price residuals on the instrument, we obtain an extremely high F-statistic of 391. This is unsurprising as the instrument matches the exact magnitude of the gas price change and as price movements on announcement days are large. Again, I apply the moving-block bootstrap of Jentsch and Lunsford (2019) for asymptotically correct inference. At the core of this approach lies the assumption that the shock of interest, the sanction intensity shock, is correlated with gas price changes on announcement days but that all other shocks in the model are uncorrelated with such announcements. This approach does not rely on changes in the shock variance as assumed in the main part of the paper. A sanction intensity shock, normalised to increase gas prices by 0.15 standard deviations as in the baseline model, leads to an increase in the spread of close to 0.1 standard deviations or 4 basis points (Figure Appendix B.11). This is exactly the magnitude as the sanction intensity shock identified by heteroskedasticity. Similarly, the effect on the stock market, on geopolitical risk, and on bond market volatility are very similar to the baseline model. These results confirm the core finding of the paper without relying on assumptions regarding the structure of the shock variances, underlining the robustness of the results.

Figure Appendix B.11: Effect of a sanction intensity shock on credit spreads, 2014 to 2024, Proxy VAR with gas price changes on sanction days

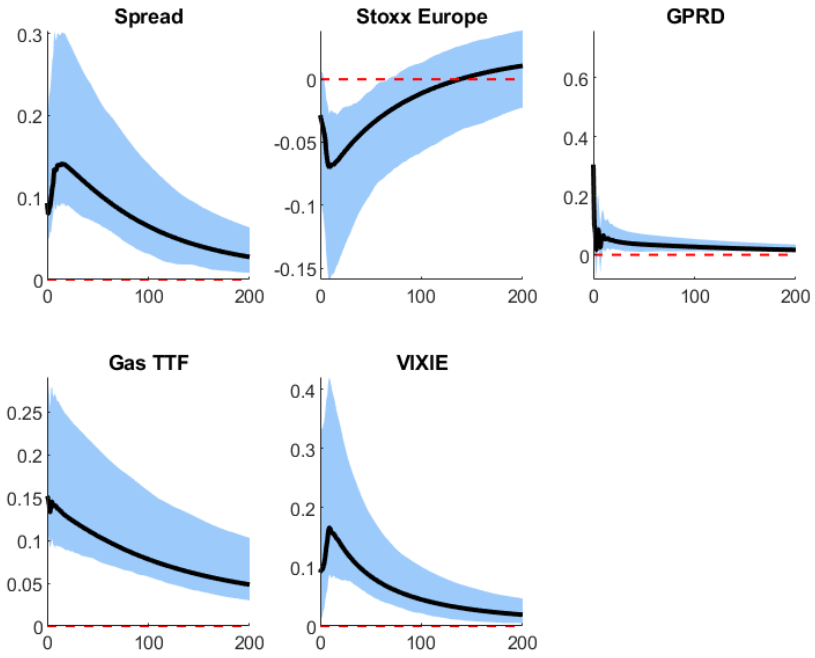


Notes: Impulse response function to an adverse sanction intensity shock together with the 90% bootstrapped confidence interval over a 200-day horizon. A logarithmic transformation is applied to the Stoxx Europe, GPRD, Gas TTF, and VIXIE. All variables are standardised to have a zero mean and a unit variance. Daily data between January 2, 2014 and April 15, 2024.

Appendix B.6 Robustness lag length and placebo tests

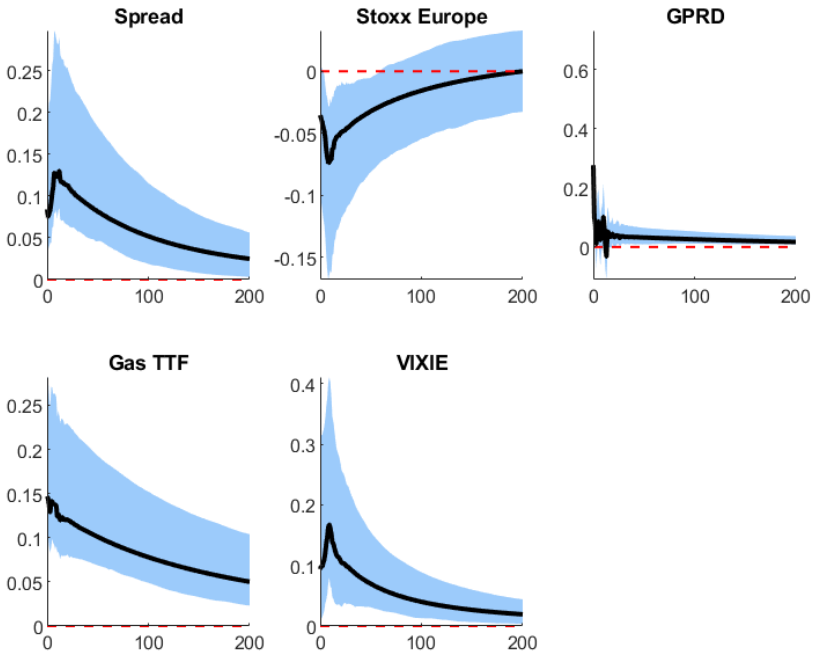
To ensure that the core result is robust to the inclusion of a higher-order of lags, I re-estimate the model with 10-day and 15-day lag lengths. Both specifications confirm the results of the baseline model. Moreover, I ensure that the results are indeed a manifestation of the selected announcement days, rather than a general feature of the chosen approach. For this purpose, I perform two placebo tests. In a first test, I randomly allocate the 44 events over the complete sample period from 2014 to 2024. In a second test, to further ensure that the results are not merely a representation of a higher frequency of events after the start of the full-scale invasion of Ukraine, I randomly scatter the same number of events separately over the pre-2022 period (19 events) and the post 2022 period (25 events) ensuring that they do not fall on actual announcement days. For each approach, I calculate the distribution of placebo responses by iteratively performing this exercise 500 times. I then validate that the original responses, obtained using the actual event days, fall outside the distribution obtained by randomly scattering events over the complete period or the sub-periods. The results, with responses for the spread, the GPR, gas prices, and bond market volatility falling outside the distribution of placebo responses, confirm that the documented effects are specific to sanction announcements and not a general feature of the data, the model, and the identification approach. The placebo distributions are centered around zero for all variables. By construction, the placebo distribution for gas prices does not contain zero on impact. For each draw, the shock is identified up to a sign only, where the sign is set to have a positive effect on gas prices. These exercises provide further validation of the selected approach.

Figure Appendix B.12: Effect of a sanction shock on credit spreads, 2014 to 2024, 10-day lag



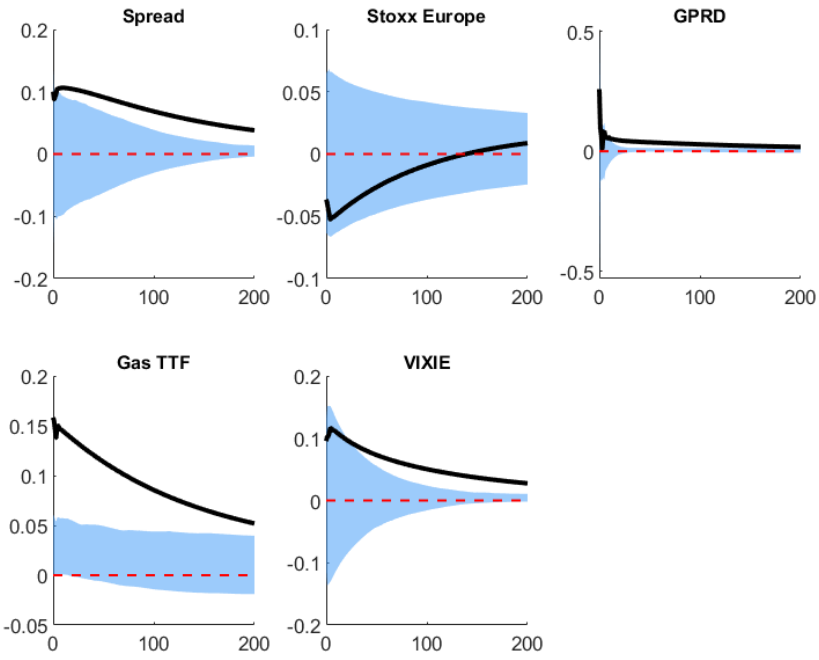
Notes: Impulse response function to a unitary adverse sanction shock together with the 90% bootstrapped standard errors over a 200-day horizon. A logarithmic transformation is applied to the Stoxx Europe, GPRD, Gas TTF, and VIXIE. All variables are standardised to have a zero mean and a unit variance. Daily data between January 2, 2014 and April 15, 2024.

Figure Appendix B.13: Effect of a sanction shock on credit spreads, 2014 to 2024, 15-day lag



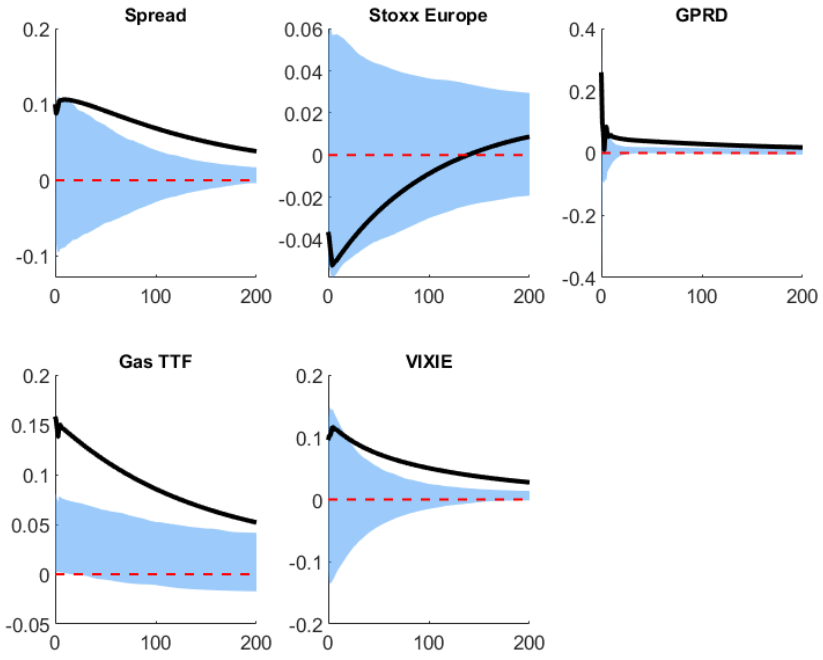
Notes: Impulse response function to a unitary adverse sanction shock together with the 90% bootstrapped standard errors over a 200-day horizon. A logarithmic transformation is applied to the Stoxx Europe, GPRD, Gas TTF, and VIXIE. All variables are standardised to have a zero mean and a unit variance. Daily data between January 2, 2014 and April 15, 2024.

Figure Appendix B.14: Effect of a sanction shock together with the placebo distribution (events randomly scattered over full sample period)



Notes: Impulse response function to a unitary adverse sanction shock together with the 90% response sample obtained from randomly scattering events over the complete sample period. All variables are standardised to have a zero mean and a unit variance. Daily data between January 2, 2014 and April 15, 2024.

Figure Appendix B.15: Effect of a sanction shock together with the placebo distribution (events proportionally scattered over the pre-2022 and post-2022 periods)

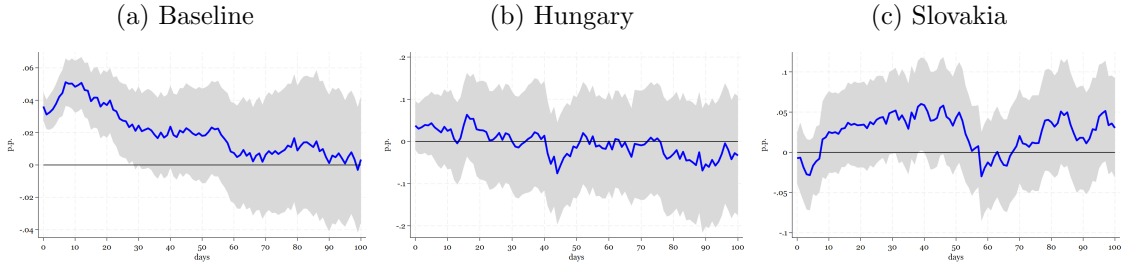


Notes: Impulse response function to a unitary adverse sanction shock together with the 90% response sample obtained from randomly scattering events over the pre-2022 and post-2022 sample periods in line with the actual count of announcements over the two sub-samples. All variables are standardised to have a zero mean and a unit variance. Daily data between January 2, 2014 and April 15, 2024.

Appendix B.7 Placebo exercise for Hungary and Slovakia

Not all member countries participated similarly in the EU's sanction regime. EU processes allowed some countries to negotiate exemptions from core sanctions and to continue importing energy from Russia. In particular, Hungary and Slovakia, blocking EU sanction measures, negotiated exemptions for oil imports and are with an import share of 80%-85% still highly dependent on Russian gas[§] (EU average 2025[¶]: 13%). We would, therefore, expect credit spreads to be less sensitive to EU-Russia sanction intensity for firms located in these countries. I estimate the model of Equation **??** for the baseline sample, as well as for the spread indices for Hungary and Slovakia separately. Bond spreads for euro-denominated bonds in Hungary and Slovakia are collected from Bloomberg. The sample includes 44 euro-denominated bonds issued by Slovakian firms and 11 by Hungarian firms. Interestingly, no significant effect of the sanction intensity shock is found for either Hungary or Slovakia on impact or after. These countries, while being similarly or even stronger exposed to the war in other respects such as geographic distance, were shielded from sanction pressures by exemptions. This exercise again confirms that the shock indeed captures sanction intensity during the EU-Russia conflict specifically, rather than merely capturing geopolitical risk more broadly.

Figure Appendix B.16: Effect of a sanction shock on the aggregate spread and selected sanction-exempted EU countries



Notes: Response of the credit spreads to a unitary sanction intensity shock together with the 90% confidence intervals. Controls include the logarithmically transformed Stoxx Europe, GPRD, Gas TTF, and VIXIE. Daily data between January 2, 2014 and April 15, 2024.

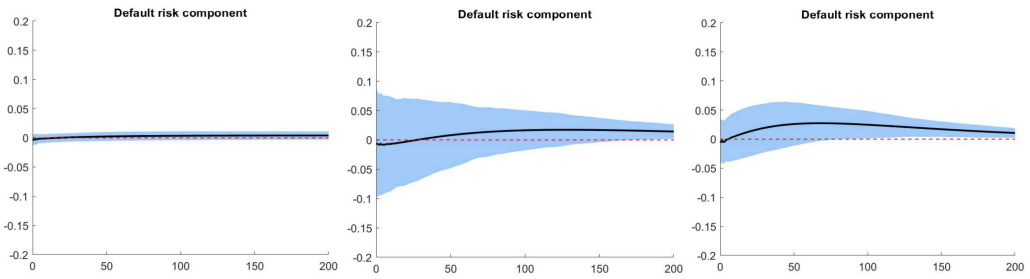
[§] Fitch
[¶] EU

Appendix Appendix C: Additional results

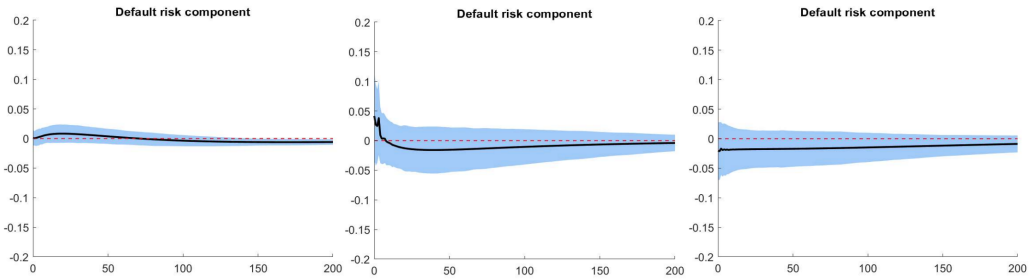
Appendix C.1 Additional results - credit market and firm data

Figure Appendix C.1: Selected industries less affected by sanction shocks

(a) Construction (b) Healthcare Products (c) Healthcare Services



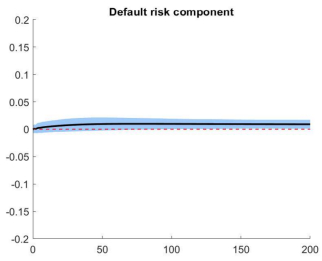
(d) Pharmaceuticals (e) Telecommunication (f) Energy - Alternate Sources



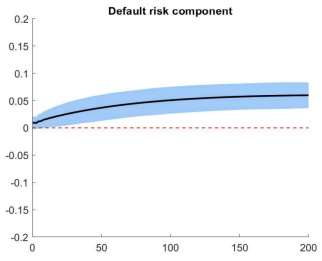
Notes: Impulse response function to an adverse sanction shock together with the 90% bootstrapped standard errors. All variables are standardised to have a zero mean and a unit variance. Only industries for which at least three different firms have bonds outstanding at each point in time are depicted.

Figure Appendix C.2: Industries - Part 1

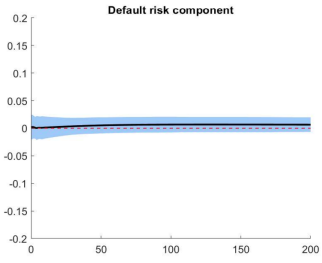
(a) Aerospace-Defense



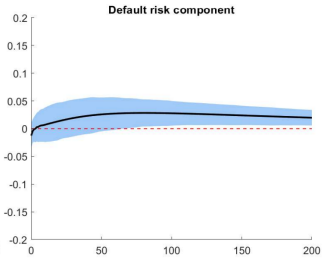
(b) Agriculture



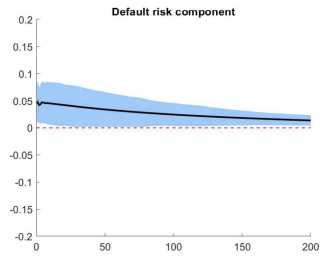
(c) Airlines



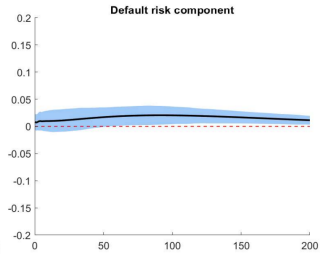
(d) Apparel



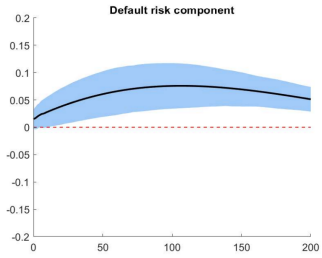
(e) Car Manufacturers



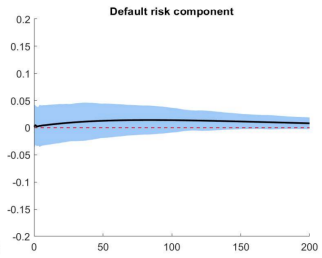
(f) Beverages



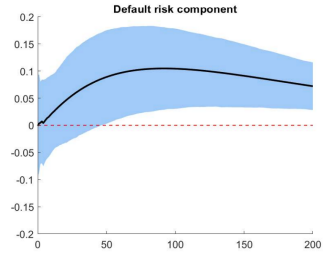
(g) Materials



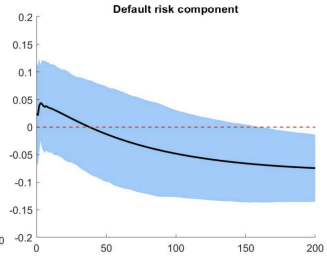
(h) Chemicals



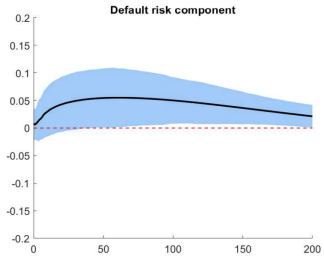
(i) Com. Services



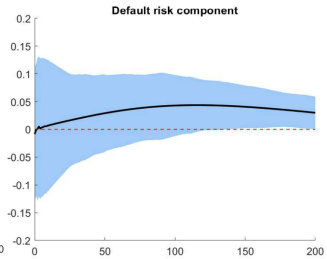
(j) Computers



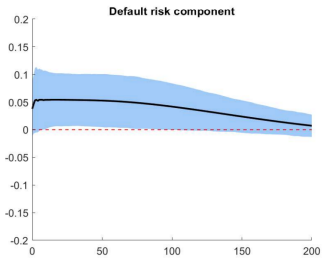
(k) Electrical Components



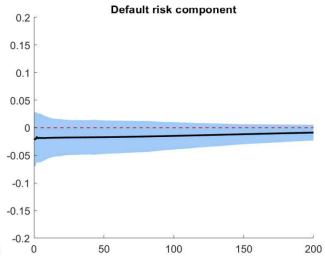
(l) Electrics



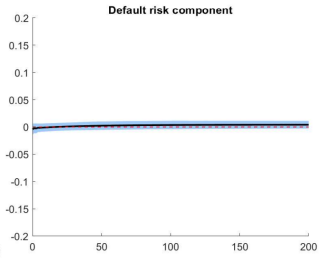
(m) Electronics



(n) Energy - Alternatives



(o) Engineering



(p) Food

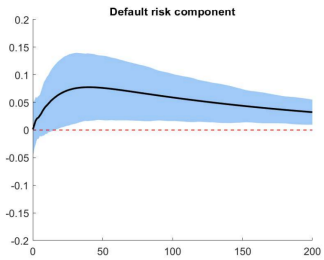
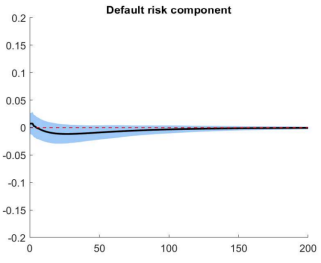
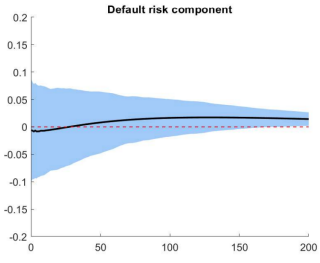


Figure Appendix C.3: Industries - Part 2

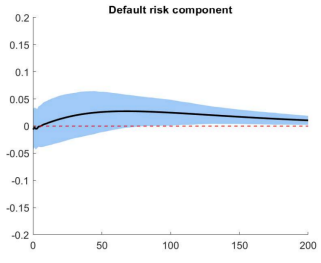
(a) Gas



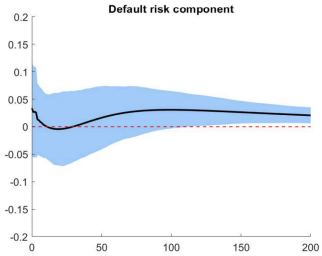
(b) Healthcare Products



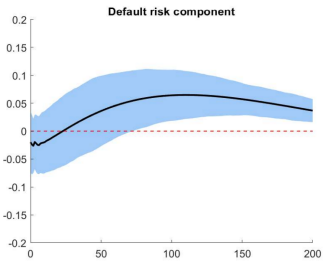
(c) Healthcare Services



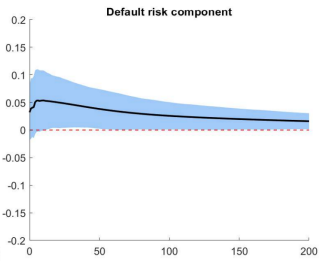
(d) Internet



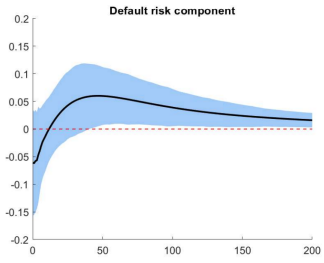
(e) Investment Companies



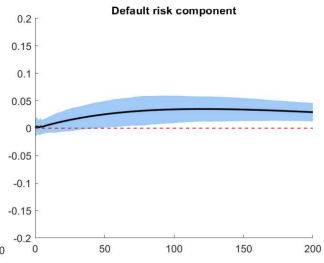
(f) Iron and Steel



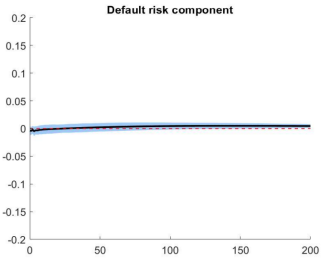
(g) Machinery



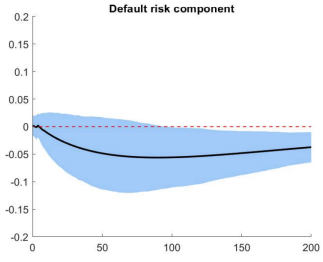
(h) Media



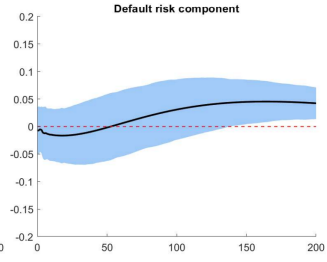
(i) Mining



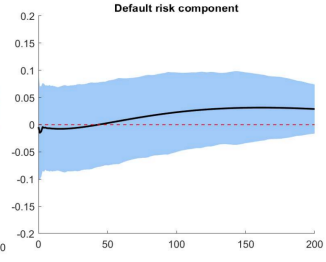
(j) Miscellaneous



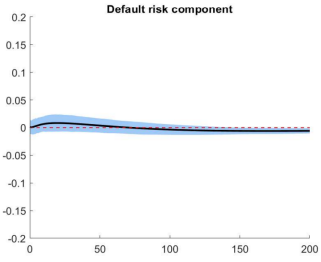
(k) Oil and Gas



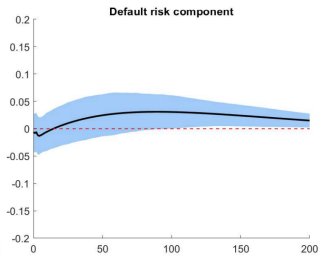
(l) Packaging



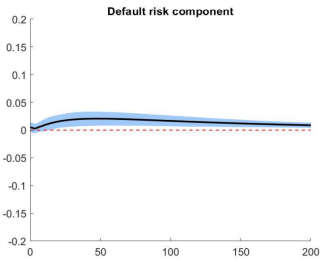
(m) Pharmaceuticals



(n) Real Estate



(o) Retail



(p) Software

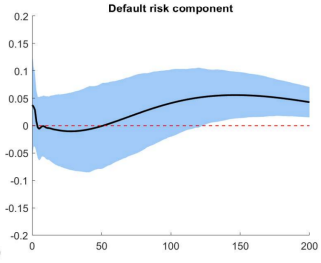
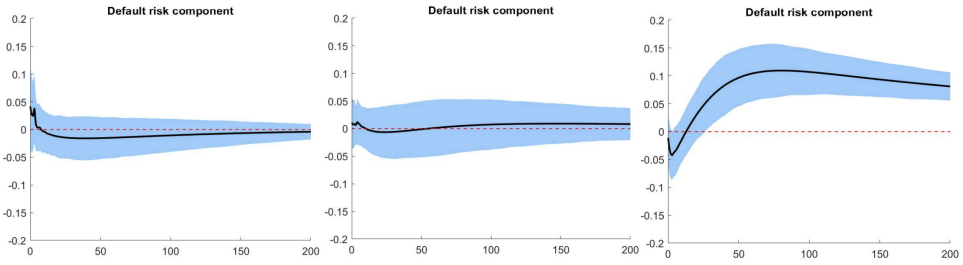


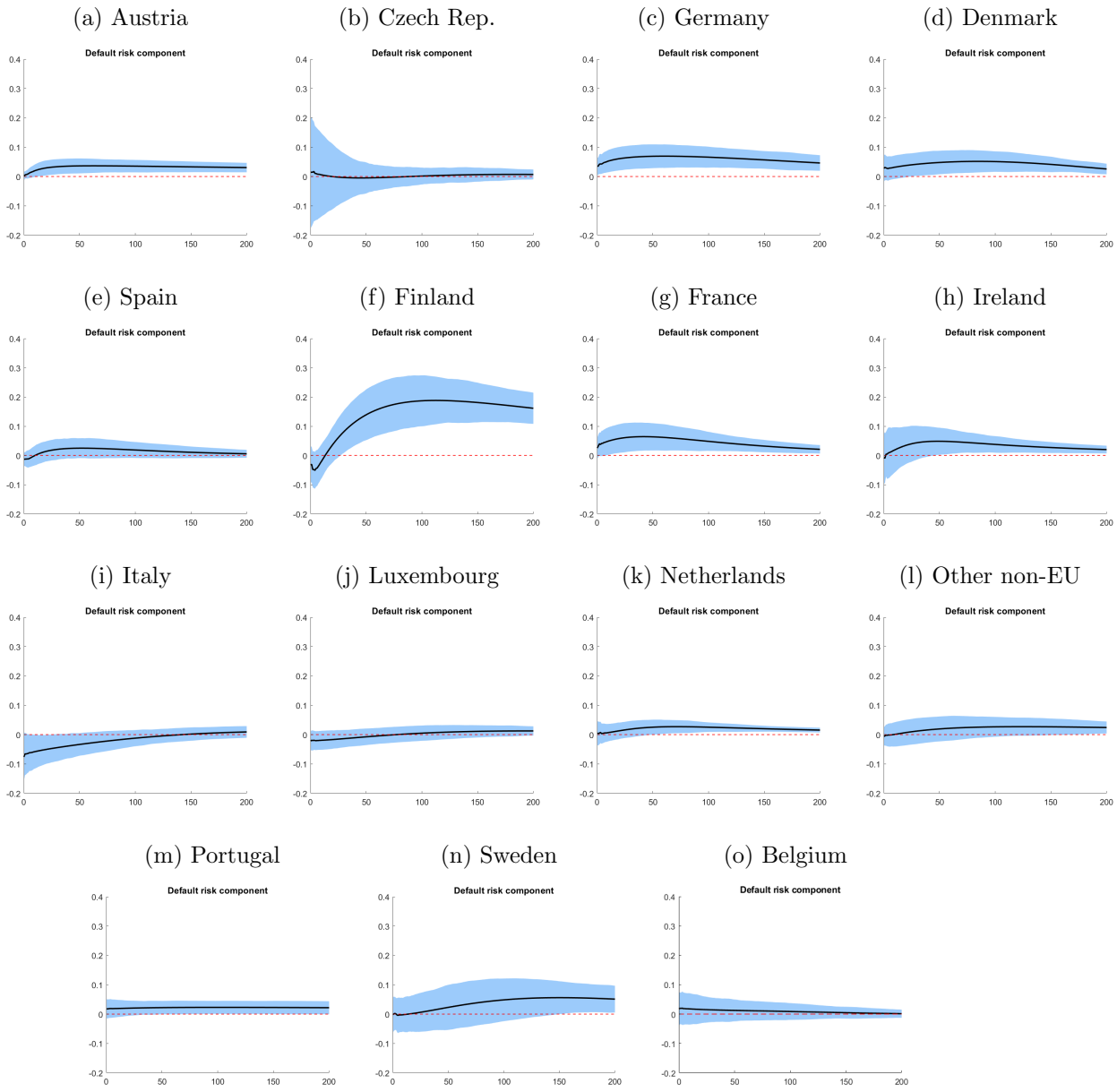
Figure Appendix C.4: Industries - Part 3

(a) Telecommunication (b) Transportation Services (c) Wood and Paper



Notes: Impulse response function to an adverse sanction shock together with the 90% bootstrapped standard errors. All variables are standardised to have a zero mean and a unit variance. Only industries for which at least three different firms have bonds outstanding at each point in time are depicted.

Figure Appendix C.5: All countries



Notes: Impulse response function to an adverse sanction shock together with the 90% bootstrapped standard errors. All variables are standardised to have a zero mean and a unit variance. Only countries for which at least three different firms have bonds outstanding at each point in time are depicted.

Table Appendix C.1: Firm financial space and conflict impact (within-firm changes)

	(1) Spread	(2) EBP	(3) Default risk	(4) Spread	(5) EBP	(6) Default risk
L.sanction_shock	0.008	0.020**	-0.012	0.043***	0.039***	0.004
	(0.016)	(0.008)	(0.010)	(0.016)	(0.007)	(0.012)
Low leverage	0.109	0.001	0.108			
	(0.230)	(0.086)	(0.214)			
High leverage	0.127	-0.030	0.157			
	(0.209)	(0.069)	(0.192)			
Low leverage*L.sanction_shock	0.012	-0.000	0.013			
	(0.019)	(0.011)	(0.011)			
High leverage*L.sanction_shock	0.080**	0.038***	0.042*			
	(0.036)	(0.013)	(0.025)			
Low EBIT				0.045	0.022	0.024
				(0.086)	(0.073)	(0.050)
High EBIT				-0.075*	-0.036	-0.039
				(0.039)	(0.024)	(0.026)
Low EBIT*L.sanction_shock				-0.001	-0.010	0.009
				(0.013)	(0.010)	(0.013)
High EBIT*L.sanction_shock				-0.013	-0.012	-0.002
N	663749	663749	663749	663749	663749	663749
Firms	189	189	189	189	189	189
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: This table examines the importance of collateral- and earnings-based borrowing constraints in a panel regression setting covering the period from January 2014 to April 2024. The dependent variables are the credit spread and its components, the excess bond premium (EBP) and the default risk component. Firms are dynamically grouped into three equally sized bins according to their leverage and earnings levels. The intermediate bin is taken as a baseline. Standard errors are clustered at the firm level.

Table Appendix C.2: Conflict impact for firms highly exposed to Russia (post 2022)

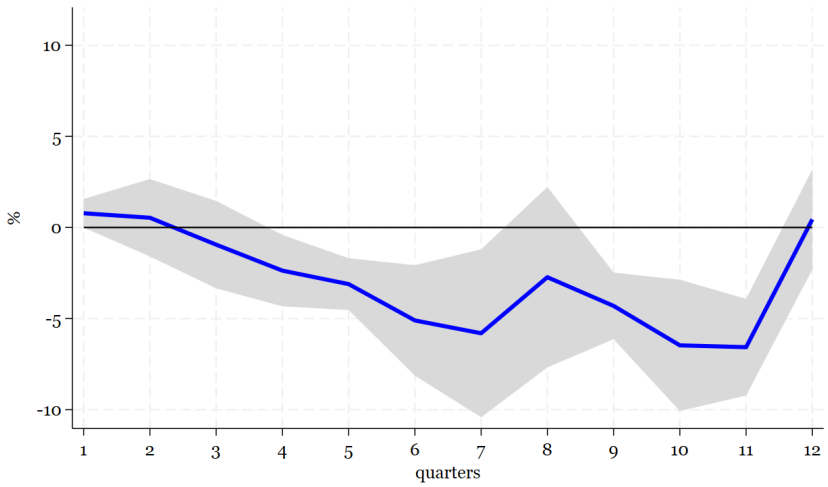
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	1-week ahead		2-weeks ahead		3-weeks ahead		4-weeks ahead	
	Spread	Spread	Spread	Spread	Spread	Spread	Spread	Spread
L.sanction_shock	0.016***	0.018***	0.021***	0.022***	0.019***	0.022***	0.018***	0.021***
	(0.002)	(0.002)	(0.002)	(0.004)	(0.003)	(0.005)	(0.004)	(0.003)
High RU risk	0.001	0.003**	0.000	-0.000	0.001	-0.003	0.002	-0.005
	(0.002)	(0.001)	(0.003)	(0.002)	(0.004)	(0.003)	(0.005)	(0.005)
High RU risk*L.sanction_shock	0.007*	0.006*	0.011**	0.007	0.011***	0.005	0.011**	0.008*
	(0.003)	(0.003)	(0.004)	(0.004)	(0.003)	(0.003)	(0.004)	(0.004)
N	266357	237908	264081	237015	260428	234799	257467	233256
Firms	160	160	160	160	160	160	160	160
Country FE	Yes		Yes		Yes		Yes	
Industry FE	Yes		Yes		Yes		Yes	
Country-Industry FE	No	Yes	No	Yes	No	Yes	No	Yes
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Controls*L.conflict_shock	No	Yes	No	Yes	No	Yes	No	Yes

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

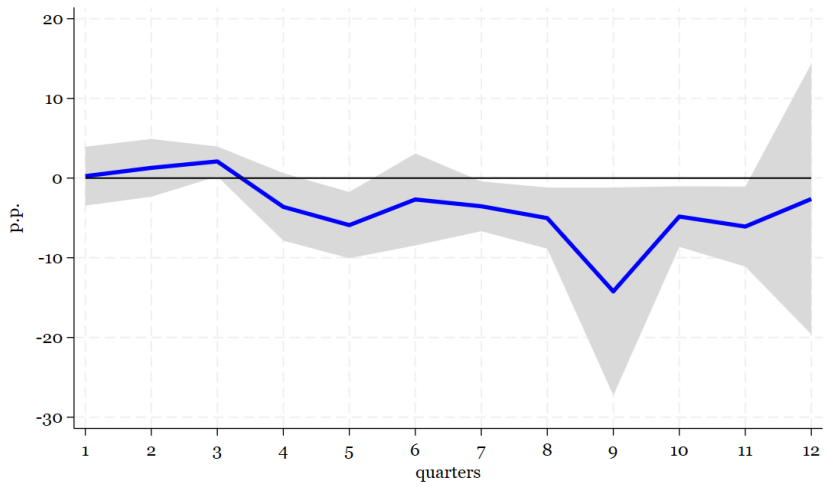
Note: This table examines the importance of firm-specific Russia dependence, measured by the index of Hassan et al. (2024), in a panel regression setting covering the period from January 2022 to April 2024. The dependent variable is the credit spread over various horizons (1, 2, 3, and 4 weeks ahead). Firms are classified as having a high Russia exposure if their Russia-specific risk index is above the 80th percentile on average in the years 2019 and 2020 (the last two available years before the full-scale invasion of Ukraine). Firm controls include total assets, EBIT, interest expenses, and leverage. Standard errors are clustered at the country level.

Figure Appendix C.6: Effect of a sanction shock on corporate investment



Notes: Impulse response function to a one standard deviation sanction shock with 90% (shaded area) confidence intervals. Standard errors are clustered at the country level. The blue line depicts the coefficient β_h in Equation 15 but including firm fixed effects. Firm controls include total assets, leverage, EBIT, interest expenses, and the average corporate credit spread (all controls are included with first-differences).

Figure Appendix C.7: Effect of a sanction shock on corporate profitability (ROE)

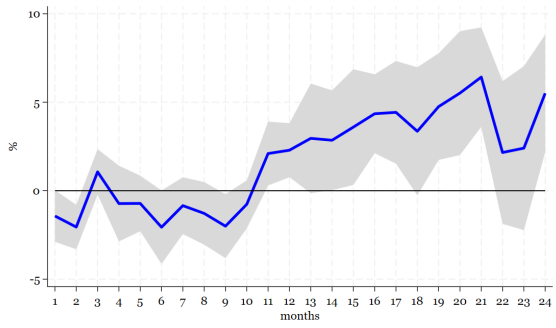


Notes: Impulse response function to a one standard deviation sanction shock with 90% (shaded area) confidence intervals. Standard errors are clustered at the country level. The blue line depicts the coefficient β^h in Equation 15 for the ROE as dependent variable. Firm controls include total assets, leverage, EBIT, interest expenses, and the average corporate credit spread (all controls are included with first-differences).

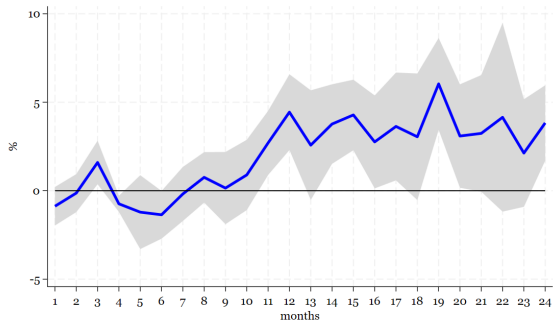
Appendix C.2 Additional results - bankruptcies

Figure Appendix C.8: Effect of a sanction shock on bankruptcy rates

(a) Control alternative shocks



(b) Excluding COVID-19 period

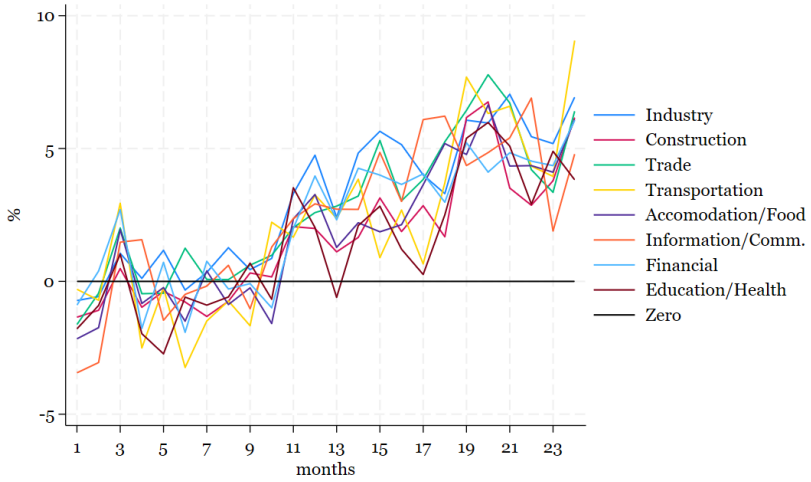


Notes: Impulse response function to a one standard deviation sanction shock with a 90% confidence interval. Standard errors are clustered at the country level. The blue line depicts the coefficient β_h in Equation 17. Country controls include the logarithmically transformed industrial production and CPI, and the 10-year sovereign bond yield and net effective exchange rate (all controls are included with first-differences). Left: Additional controls are the European EPU index, the GPR index, the EU gas price shocks of Alessandri and Gazzani (2025), and EA monetary policy shocks of Jarociński and Karadi (2020). Right: Sample period excludes January 2020 to September 2021.

Only 9 EU countries report bankruptcy data on a monthly frequency, whereas data for 18 EU member countries is available quarterly. To verify the results presented in Figures 10 and C9, I re-estimate the local projections with one-quarter lags using the full sample of quarterly available bankruptcy data. The results confirm that bankruptcy numbers rise after one year and increase further after (Figure Appendix C.10). The estimated cumulative response after two years is somewhat stronger than in the monthly model.

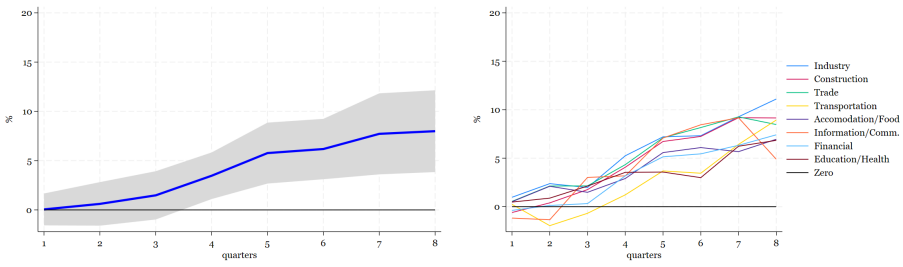
Lastly, I estimate the model to analyse the effect of the sanction intensity shock on monthly trade with Russia using UN Comtrade data. As expected, aggregate export and import volumes drop drastically after increased conflict. The cumulative monthly reduction in trade reaches up to around 10% of the average monthly import volumes from Russia and 12% of the average export volumes to Russia between January 2014 and April 2024. On the import side, the largest reduction is by far observed for mineral fuel and oil products (Figure Appendix C.11). However, also other commodities imported from Russia are affected by the sanction shock. On the export side, the reduction in trade is strongest for machinery products, electrical equipment and machinery, and vehicles (Figure Appendix C.12). These sectors all experienced a significant increase in the default risk component.

Figure Appendix C.9: Effect of a sanction shock on bankruptcies



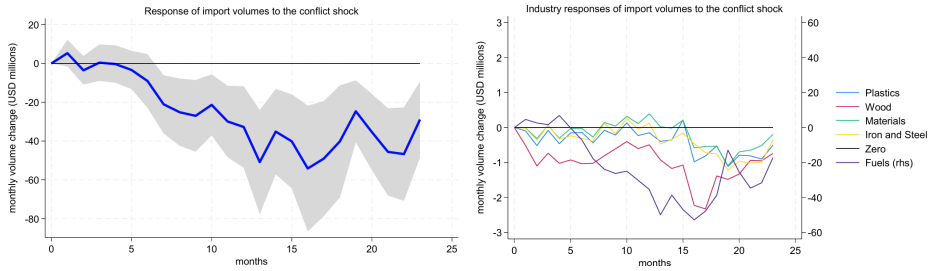
Notes: Impulse response function to an adverse sanction shock. The blue line depicts the coefficient β_h in equation 17. Country controls include the logarithmically transformed industrial production and CPI and the 10-year sovereign bond yield and net effective exchange rate (all controls are included with first-differences). Significant responses are observed for all industries except Education/Health on a 10% significance level after two years.

Figure Appendix C.10: Effect of a sanction shock on bankruptcies (quarterly data)



Notes: Impulse response function to an adverse sanction shock with a 90% confidence interval. Standard errors are clustered at the country level.

Figure Appendix C.11: Effect of a sanction intensity shock on EU countries' import volumes from Russia



Notes: Impulse response function to an adverse sanction shock with a 90% confidence interval. Monthly trade data between the EU and Russia from 2014 to 2024 is taken from UN Comtrade. Standard errors are clustered at the country level.

Figure Appendix C.12: Effect of a sanction intensity shock on EU countries' export volumes to Russia



Notes: Impulse response function to the sanction shock with a 90% confidence interval. Monthly trade data between the EU and Russia from 2014 to 2024 is taken from UN Comtrade. Standard errors are clustered at the country level.

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